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Evaluation methods of port dominance: A critical review

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ABSTRACT

With the continuous advancement of the global economy and international trade, the competition among ports becomes increasingly fierce. Numerous studies show that evaluation methods of port dominance can effectively quantify the appropriate dominance index and find the weak links to enhance ports' comprehensive competitiveness. This paper summarizes the port dominance evaluation methods in the existing literature and expounds their importance based on the development of information technology in shipping activities. Moreover, the application scope, the advantages, and disadvantages of current evaluation methods are explained using qualitative, quantitative, deductive, and retrospective methods. It was found that, with the development of information technology, evaluation methods are becoming more comprehensive, targeted, specialized, refined, digitalized, and intelligentized. It was also found that since existing studies often adopt single-faceted evaluation methods and select a single factor for evaluation, they cannot effectively analyze and evaluate the real advantages of the port. In addition, the present paper discusses the prospects of ports and potential research.

1. Introduction

China, as the world's main "commodity processing plant", is responsible for serving most countries and regions in the world, including Europe and North America (Wang and Liu, 2013). The Ministry of Transport of the People's Republic of China issued the Bulletin of the 2019 Transport Development Industry. The data show that the number of berths in ports of 10,000-ton class and above still has the potential for continuous growth in recent years. Moreover, many ports continue to integrate together to overcome the problems of overcapacity and vicious competition. As of the end of 2019, in ports across China, there were 22,893 berths, of which 2520 berths were the 10,000-ton class and above. Note that this number shows an increase of 76 berths compared with 2018. In addition, the proportion of berths in the 10, 000-ton class and above rapidly increased from 6.7% in 2014 to 11.0% in 2019. However, China's investment in coastal construction has been decreasing since 2013 and the overall growth rate of all kinds of new berths is declining every year, as per another set of data. In addition, this competition has intensified because emerging transshipment centers and other gateway ports compete with traditional hub ports. It is worth mentioning that, in 2020, due to the recession of global economy, the foreign trade pressure continues to expand, especially under the dual influence of trade friction and COVID-19, resulting in the poor operation of the global supply chain and the reduction of the demand for international trade.

Dominance is an index to measure the status and role of a certain factor or aspect in the port group. Port competitiveness reflects the comprehensive ability, which is influenced by multiple strong dominances. To promote the survival and development of port enterprises and optimize the efficiency of port service, it is necessary to use reasonable evaluation methods to correctly assess the port competitiveness and determine its advantages. Thus, the effective allocation of resources is realized, and the comprehensive competitiveness is improved. To this end, this paper systematically summarizes the evaluation methods of port dominance and expounds their importance in terms of developing information technology in shipping activities. Moreover, the application scope, the advantages, and disadvantages of current evaluation methods are explained using qualitative, quantitative, deductive, and retrospective methods to provide a quantitative basis for the transformation and upgradation of ports (Figs. 1 and 2).

2. Background of port dominance

An earlier study investigated the port competitiveness from the economic perspective. In this study, the port was assumed as a part of the shipping economy, and the focus was on the role of ports and

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shipping in economic activities (Ferretti et al., 2018; Frankel, 1987; Krogsgaard et al., 2018; Liu et al., 2014; Yin et al., 2011; Robinson, 2002). In recent years, China has taken "Maritime Power", "Polar Silk Road" (Liu et al., 2020) and "the Belt and Road Initiative" as important strategies in the new era. Thus, ports have played an increasingly prominent role in national security and geopolitics, and they control the transportation of strategic materials and monitor key shipping routes. Therefore, the port's strategic development and construction planning are important for China to highlight its power (Brian and Philip, 1999) (Fig. 3).

It is worth mentioning that a port is an external window of a region or city (Jitendr et al., 2012; Kong and Liu, 2021). Thus, for evaluating the port dominance from the perspective of its own conditions and accessibility to the hinterland, natural and objective factors such as geographical location (Malchow and Kanafani, 2001; Tiwari et al., 2003; Ren et al., 2018), port infrastructure (Jeevan et al., 2019; Van Dyck and Ismael, 2015), transportation capacity (Song et al., 2016; Tovar et al., 2015), urban influence (Kim, 2014; Yeo et al., 2014), and strategic hub (Calatayud et al., 2017) should be considered. From the perspective of sea-land intermodal transport, Mou et al. (2018) comprehensively analyzed five factors, including road network density, and influence degrees of traffic trunk lines, cities, shipping strategic hubs and shipping strategic channels. They also analyzed the location dominance of important ports along the "Maritime Silk Road". Peng (2013) indicated that the ports of Zhoushan Islands, based on their spatial location conditions and regional factors such as throughput capacity, collection and distribution system, operation and service network, form the regional attraction to hinterland goods. Li (2019) used five factors, including the port location environment, infrastructure construction, logistics capacity, urban economic strength, and factor input, to construct the comprehensive competitiveness evaluation index system of major ports in the Yangtze River Economic Belt. Li et al. (2016) selected 11 individual indices from five perspectives of natural conditions, infrastructure, production conditions, policy support, and economic environment to build a comprehensive evaluation index system for port competitiveness. Wu et al. (2020) analyzed the development data of ports and their locations. They indicated that there is an interactive effect between regional ports and economic development. In addition, they showed that the contribution of ports to economic development is stronger than the reverse effect of the economy. To conclude, the above-mentioned evaluation methods are only based on the port conditions, failing to consider the attraction to offshore vessels and relevant service levels after the port is put into operation. Therefore, it cannot provide time-varying evaluation indices.

With the advancement of big data technology, especially the wide application of automatic identification system (AIS) technology in the field of marine trade, the development of port dominance evaluation has been realized from the simple port facilities to the comprehensive analysis combined with the actual operation efficiency. By objectively analyzing the composition of port throughput and logistics statistics, the development environment and port competitiveness have been analyzed in different studies (Hao et al., 2015; Kim, 2016; Zhu and Cai, 2018; Yang and Wang, 2016). Zhao and Zhou (2019) and Da Cruz and De Matos Ferreira (2016) suggested to quantize the port throughput, scale, city and hinterland, and future development planning so that the data can be used to construct an evaluation system of port advantages around Bohai Sea. Xu (2018) used a ship's AIS data to analyze the turnaround time of a ship in port, residence time of a ship at berth, berth occupancy, transportation capacity and traffic flow of cargo, which were the evaluation indices of port operation efficiency for analyzing port advantages. With the diversification of port development, port competitiveness has gradually evolved into the competition of comprehensive strength. Note that, with the combination of hard and soft power, the influencing factors are becoming more complex and diverse. Based on previous studies, Gong (2015), Yang (2012), Lv and Zhang (2018), Qu et al. (2014), and Teng and Hu (2017) extracted individual variables representing all information and evaluated port competitiveness by using the principal component analysis (PCA).

In recent years, with the diversified development of ports, the factors affecting the port competitiveness are more complex. Note that the future traffic flow of ports, which can be analyzed using computer simulation technology, is taken as an important condition to evaluate port dominance. The simulation and prediction methods have gradually been applied in the evaluation of port dominance. Duan et al. (2015) adopted the $\mathrm{UTA}^{\mathrm{GMS}}$ method, combined with extreme ranking analysis (ERA) and Monte Carlo simulation method, to conduct an empirical analysis on the competitiveness of 16 inland river ports in the Yangtze River trunk line of China. The list ranking from the best to the worst port along the Yangtze River and the full reference sequence of all ports were given. Peng et al. (2018) designed a comprehensive evaluation model, called CCPE (conditions, capacity, potential, and efficiency) model, to measure port competitiveness through 18 factors related to conditions, capacity, potential, and efficiency. This model was based on the data of geographical environment, freighter trajectory, port infrastructure, and regional socio-economic aspects.

Complex systems science is widely used in the Internet, social

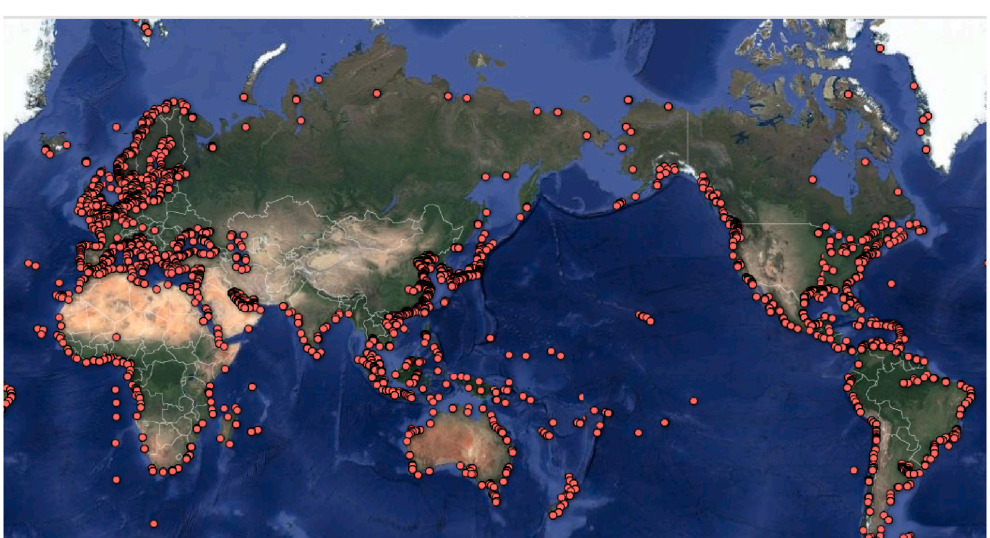

Fig. 1. Distribution of major ports in the world.

networks, disease epidemics, and even protein medicine (Boccaletti et al., 2006). As the characteristics of the shipping network are in line with complex systems science (Ducruet, 2015), this science technology can evaluate port dominance by using ports as nodes and routes as paths. Thus, the topological relationships, port groups, and network communities in the shipping network can be analyzed (Fig. 4). Note that, during navigation, different types of ships have different shipping network topologies. For example, the tanker shipping network presents the characteristics of the shortest route for economic optimization (Fig. 5), and the fishing trawler shipping network has cash distribution characteristics (Fig. 6). In addition, the degree (Dorogovtsev and Mendes, 2004), the clustering coefficient (Newman, 2003), and other key indices are used to find the key elements that affect port dominance. Wu et al. (2019) developed a complex network, in which important ports along the Maritime Silk Road were used as nodes. They comprehensively analyzed the characteristics of the network's average path length, clustering coefficient, degree, and distribution to evaluate the importance of ports along the Maritime Silk Road. Wu (2020) designed a topological map from a complex network of the port groups in Jiangsu Province and analyzed the topological structure. In this study, indices such as the degree, the betweenness centrality, and the clustering coefficient of nodes in the network were calculated. In addition, the current status, influence, and development potential of Jiangsu ports were calculated and analyzed based on these indices. Based on the multiple flow theory, Li and Li (2019) analyzed the competitive pattern of ports by using significant flows in the transportation network. Xu et al. (2021) used a game-theoretical model to analyze the co-opetition between two ports with sharing capacity. They discussed three competitive game interactions, called benchmark model, passive sharing, and proactive sharing, to maximize the payoff by determining the optimal berth quantity and service price.

3. Evaluation methods of port dominance

The research on the evaluation of port dominance has extended from simple economy and trade to a stage involving diversified and complex factors such as peripheral security and geopolitics. The evaluation methods have developed from subjective and objective qualitative evaluation to quantitative evaluation, simulation, and retrospective inference. Diversified and advanced evaluation methods are used to make a reasonable evaluation of port dominance. The port is a complex system involving many fields, disciplines, and influencing factors. Therefore, the present paper systematically summarizes the studies of port dominance evaluation methods in recent years.

3.1. Literature review of qualitative evaluation methods

In the early stage of port development, its functional structure is single, limited by technical and geographical conditions. Note that the port's geographical regional advantage plays a decisive role. With further development of human production activities, more developed coastal areas have established trade ports. Thus, ports and cities present the trend of integrated development, and both interact with each other. After the third industrial revolution, the development of roads, railways, and ships as well as the growth of international trades have promoted the continuous expansion of ports. Furthermore, ports have been served for various functions, including freight, carrying passengers, travel (Chen et al., 2016), and logistics (Chen et al., 2014a). The Age of Discovery has shifted human development to the sea so that the ports in straits, canals, and key channels have a prominently political and strategic status. There appeared a large number of countries and cities rising from ports and living in ports. These ports have become the battleground for great powers to play games and safeguard national strategic security.

The literature review of qualitative evaluation methods is made based on observation and analysis of the development background, reality, and state of ports. Evaluation methods mainly include:

1. Expert analysis: It takes experts as the object to obtain information and makes personal judgment with the help of experts' knowledge and experience.
2. Benchmarking analysis: By comparison, the port with strong competitiveness is selected as the benchmark, and the advantages of the benchmark port are compared and analyzed. Then, the methods to narrow the gap and improve the competitiveness are proposed.
3. Technique for order of preference by similarity to ideal solution (TOPSIS) method (Hwang and Yoon, 1981; Li, 2014): the judgment matrix is established by taking the factors that affect port dominance as indices. Then, after a series of processing, positive and negative ideal solutions are obtained, and the Euclidean distance and relative proximity between the target values are calculated. Finally, the relative proximity degree is used to rank the port dominance.
4. Hierarchical analysis (Xu, 1988): According to different evaluation dimensions and evaluation index systems, port dominance is divided into several criteria layers, which are evaluated layer by layer, and finally the total ranking is obtained.
5. Fuzzy comprehensive evaluation method (Liu, 2011): A fuzzy judgment matrix is built to make a comprehensive evaluation of uncertain factors in evaluating port dominance using the fuzzy mathematical method.

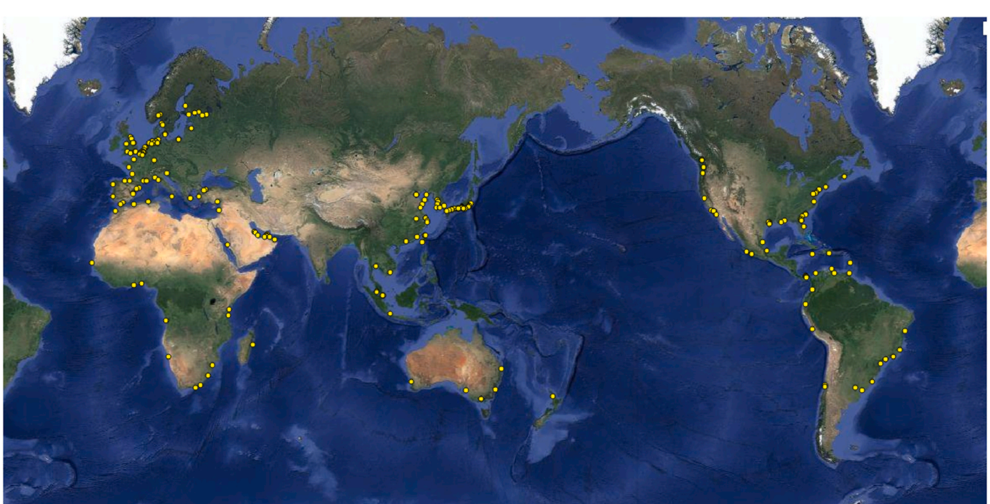

Fig. 2. Distribution of key strategic ports in the world.

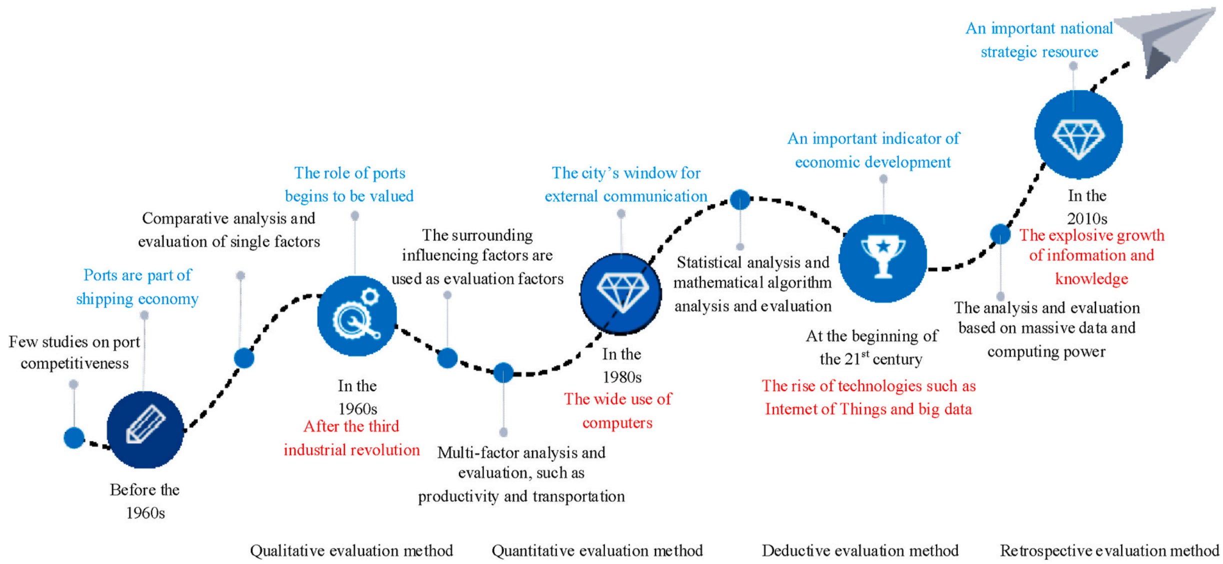

Fig. 3. Development of port evaluation methods.

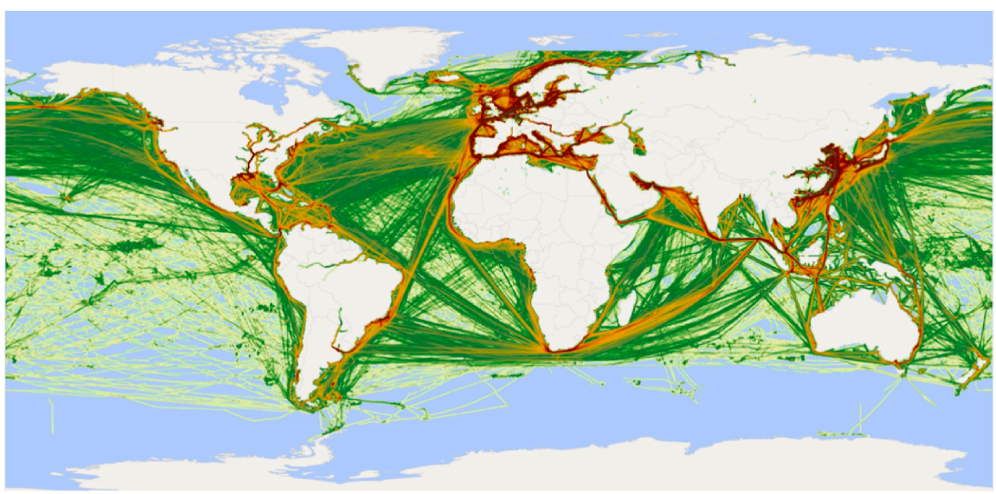

Fig. 4. Shipping network of all ships in the world within one year.

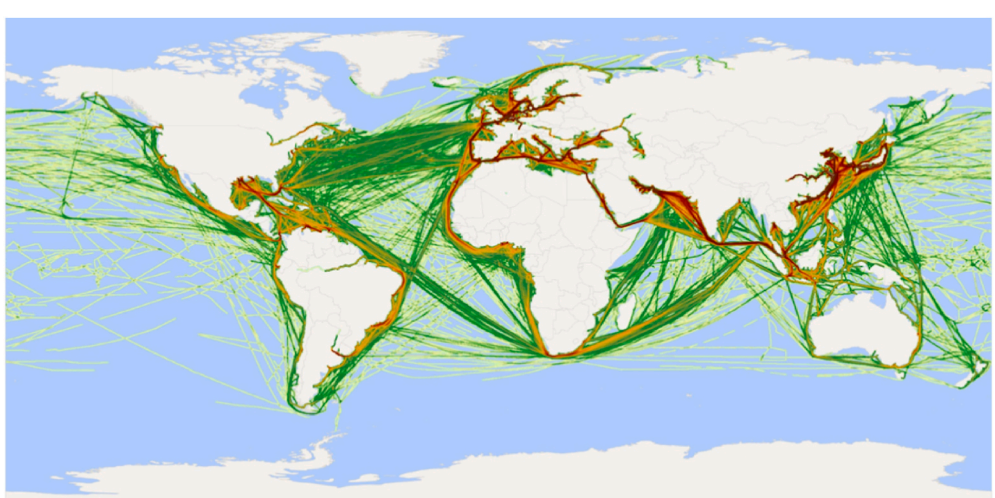

Fig. 5. Shipping network of very large crude carriers in the world within one year.

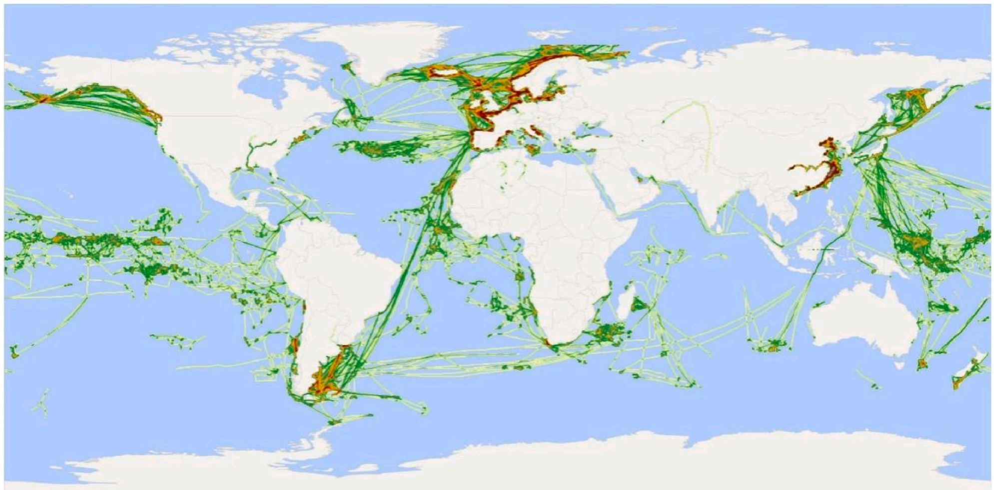

Fig. 6. Shipping network of fishing trawlers in the world within one year.

3.2. Quantitative evaluation methods based on mathematical statistics

With the wide application of computer technology in shipping, the construction of shipping informatization increases the information data, which are sufficient for the evaluation of port dominance. Note that the previous qualitative evaluation methods fail to meet the needs of port evaluation. Quantitative evaluation is to collect and process data and make a quantitative judgment of port dominance with mathematical statistics. The evaluation data mainly include port throughput, operation and management efficiency, berth data of ships, shipping trade data, urban development data, and economic investment data (Xie et al., 2021).

Quantitative evaluation methods based on mathematical statistics reflect the facts objectively, including the entropy weight method, factor analysis, principal component analysis, and data envelope approach (Yang and Wang, 2016; Li et al., 2007; Zheng and Xu, 2010; Wang and Cheng, 2010).

1. Entropy weight method (Wan et al., 2021): The entropy weight is used as the weight value, which is a weight determination method of the information content of influencing factors of port dominance. If the amount of useful information is little and the uncertainty is high, its entropy value is large and has a small weight value. Otherwise, the entropy value is small, and the weight value is large. This method can be used to objectively evaluate the weight of many factors affecting port dominance.
2. Factor analysis (Gorsuch, 1983): Factor analysis and principal component analysis belong to a class of methods that use fewer factors to link more indices or factors. The factors greatly affecting port dominance are fused into one factor, and only several factors are used to analyze and evaluate port dominance finally.
3. Data envelope approach: It is a nonparametric statistical evaluation method. The weights of input and output indices are used as variables to dynamically adjust the weights of the evaluation model. Then, the evaluation is performed based on the most influential factor. By using the optimal weight to calculate the efficiency value of other influencing factors, the most critical factors affecting the port comprehensive efficiency can be found.

3.3. Evaluation methods based on simulation and deduction

With the rapid development of information technology, the Internet of Things, supercomputing, big data, and virtual reality are constantly being applied in the shipping industry. Thus, information technology results in the revolutionary development of port informationization so that data have gradually become important assets of port enterprises. In addition, accumulating data for years provide resources for port planning and future development prediction. With the help of big data analysis and prediction as well as simulation and deduction technology, the problem of blindness in port planning and development direction is effectively solved, and a more accurate, intuitive, and scientific method is provided to evaluate port dominance. In recent years, experts have applied computer simulation and prediction to the research of port construction, environmental adaptability, development strategies, and navigation channel mapping, and possible shortcomings in the future development of ports have been found. The main methods include simulation and deduction, big data analysis and prediction, and back propagation (BP) neural network.

3.3.1. Simulation and deduction method

The simulation and deduction method is to find out the significant nonlinear characteristics through the statistics of ship traffic flow data, and to deduce the future development trend through the modeling of historical data. The traditional deductive methods include: K-nearest neighbor regression model (Lv, 2011), time series model (Kuang and Huang, 2004), grey model (Li et al., 2004; Zheng and Xu, 2011), and probabilistic directed graph model (Chen et al., 2020). However, due to different advantages and disadvantages of a single model, some researchers combine them together. Thus, the advantages of each model can be kept, and the corresponding disadvantages can be avoided (Lv and Zhang, 2018; Fan and He, 2008; Dia, 2001; Gan and Chen, 2006; Rao et al., 2007), and the accuracy of deduction can be effectively improved. With combined models, the deduction results can reflect the time-varying and nonlinear characteristics of the data effectively and express the data patterns correctly.

3.3.2. Big data analysis and prediction method

With the deepening of port enterprises' cognition of the concept of big data, the collection and in-depth mining of shipping data are playing an increasingly important role in the fine management of shipping (Sun, 2020). Port enterprises have been building big data centers to collect all kinds of data generated by daily port production in real time (Xu et al., 2021). According to the port production management logic, the big data analysis model is constructed to predict the future development trend of port throughput, logistics management, and process management, which can dynamically supervise the port production process and improve the daily operation efficiency of the port. With the help of big data analysis, the internal relationship between the data can be mined. By analyzing abnormal data, such as short-term port congestion, too high berth ratio, and safety accidents, the model is able to find the cause intelligently. Thus, functions like safety management and emergency warning can be achieved, and the quality of port service can be improved (Xing, 2019).

3.3.3. Neural network reasoning method

With the rise of artificial intelligence technology, neural network has become the main research content of port throughput modeling and reasoning (Yang et al., 2020). The main models include BP, radial basis function (RBF), and Elman neural networks (Yang and Wang, 2016; Yang and Cui, 2012). Note that the BP neural network has the function of error feedback, thus it is more effective to model and reason the port throughput. However, in practical applications, the main influencing factor is the neural network parameter. Therefore, in order to improve the accuracy of modeling and reasoning, many researchers have introduced other algorithms to determine the parameters of the neural network (Chen and Lan, 2019), or improved and optimized the neural network (Sun, 2002; Zhao et al., 2020; Chen, 2020; Huang et al., 2020; Wang et al., 2020; Huang and Chen, 2019), so as to improve the prediction accuracy.

3.4. Retrospective evaluation methods based on relationship mining

Port competitiveness is a form of comprehensive evaluation of ports, which is the result of a variety of complex factors, and there are countless correlations between various factors. Thus, port competitiveness reflects the advantages and disadvantages of ports in some aspects. Therefore, it is difficult to find out the internal relations and influencing reasons of the factors influencing port dominance through the mere analysis of several main factors. The retrospective evaluation method, based on the shipping network relationship, speculates the cause or mechanism of the phenomenon by using the characteristics of port dominance in some aspects. The retrospective evaluation method has become the most important method for evaluating port dominance. It mainly includes the Bayesian network model and complex network analysis.

3.4.1. Bayesian network model (Zhang and Wang, 2020)

Bayesian network model is a mathematical model based on probabilistic reasoning, which uses known variables to predict the probability of unknown variables. It is one of the most effective models in the field of uncertain knowledge expression and reasoning. In this model, ports are represented as nodes of variables, routes connecting these port nodes are represented as directed edges, and the relationship strength is expressed by conditional probability. Bayesian network model is used to quantitatively express the interaction relationship between the factors that affect the variables and the results through forward and reverse reasoning. Based on the results, the degree of shipping patency between ports and the degree of trade intimacy between ports can be accurately calculated. In addition, through probabilistic statistical modeling, the high-risk factors that affect port operation efficiency and shipping patency rate are analyzed.

3.4.2. Complex network analysis (Watts and Strogatz, 1998)

The structure of complex systems and its relationship with system functions have always been hot topics in the scientific field. The complex network analysis is applied under the following conditions. 1) there are a huge number of nodes. 2) the network structure presents many different characteristics. 3) there are differences in connection weights between nodes and there is a possibility of directionality. 4) nodes and connections have complex changes over time and interact with each other (Hofman et al., 2017). Taking ports as nodes and routes as the connection of nodes, Fleming and Hayuth (1994) used location attribute indices for the first time to demonstrate the strategic role of each port. Notteboom and Rodrigue (2005), by introducing complex network indices, evaluated the central role of ports in shipping networks. In addition, Chen and Hu (2016) found that judging the role and function of ports by the measurement index of centrality helps to find the port nodes that play a key role in the structure and function of the shipping network. By representing the degree of connection between ports with centrality, the impact of ports on the function of shipping networks with vulnerability, Wu et al. (2019) demonstrated the role of ports in the region from a different perspective.

3.4.3. Topological structure analysis

Transportation network is a directional, weighted complex network (Li et al., 2017). Therefore, the shipping network has unique topological structure characteristics, and the choice of routes will cause a great difference to the network structure characteristics (Shibasaki et al., 2017). To maximize profits, shipping enterprises will comprehensively consider the constraints of port construction, geopolitics, taxation, route weather, and hydrometeorology on the cargo transportation destination when selecting service routes or arranging sailing lists. Note that a slight fluctuation of the above factors affects the shipping route and causes a certain change in the network's overall topology. The topology of shipping network is mainly concerned with the connectivity and robustness of the network. Pan and Wang (2017) analyzed the port connectivity using complex networks and selected six features of network topology as the comprehensive evaluation indicators of port connectivity. These features include the degree centrality, betweenness centrality, eigenvector centrality, coreness, clustering coefficient, and average path length. Based on the traditional complex network, Li et al. (2017) constructed the weighted complex network to reveal the topological structure characteristics of the port shipping network, and selected six network topological characteristics, including degree, node strength, average shortest path length, clustering coefficient, matching coefficient, and network structure entropy as the port evaluation indices. They hierarchically divided the port nodes in the network using cluster analysis. Wu et al. (2017 and 2018) measured the network's robustness using the average degree, the proportion of isolated nodes, and the clustering coefficient. Peng et al. (2017) simulated the attack strategy of port node degree value and betweenness centrality value in the shipping network, and found the failure threshold of the shipping network and the key global ports. Achurra-Gonzalez et al. (2019) used the game theory to build an offense and defense simulation model, and established the anti-interference level and port failure level of the shipping network under different damage degrees and diversion strategies. Liu and Hu (2016) used two simulation methods, including random attacks and deliberate attacks, to evaluate ports from the perspective of the connectivity and functional robustness.

3.4.4. Community structure analysis

The community is a collection of ports with similar attributes in the shipping network. The ports in the same community are closely related, while those in different communities are independent. The main methods for dividing communities are:

1. The traditional community division is based on geographical locations, such as continents, countries and regions.
2. Shipping routes are used as links, and the closeness between ports is used as the standard for dividing communities. This method can be exemplified by the ports along "the Belt and Road Initiative" and domestic coastal ports (Chen et al., 2014b), and ports along the Yangtze River (Chen et al., 2015a,b) and Bohai Bay Rim (Chen and Lu, 2012).
3. Communities are divided according to the types of maritime activities such as container ships, coal carriers, crude oil carriers, iron ore carriers, and cruise ships (Lu et al., 2015).

Note that the studies on community structure facilitate the analysis of different roles of ports or port groups (Xu et al., 2021) in global networks and communities. There is usually one or two core ports in the community with strong comprehensive competitiveness, and the ports have complementary advantages, which can be used as a method to evaluate port dominance in the region. Girvan and Newman (2001) proposed the well-known modularity as a metric to qualitatively analyze the network division, and to quantitatively describe the communities in the network. Jiang et al. (2018) constructed a Newman fast algorithm based on modularity to calculate the Maritime Silk Road shipping network. It was found that several core ports and communities have the leading position, and their degree follows the power-law distribution, while the other communities are small. To improve the dispatching and operation efficiency of the ports in the community, Zheng et al. (2019) studied their transportation efficiency. Pan et al. (2019) proposed an effective method to identify major port communities and to analyze global shipping networks' connectivity based on the community structure.

3.5. Summary

By analyzing and sorting out the research results of port dominance evaluation, it was found that with the development of advanced information technology, the methods and means of dominance evaluation are becoming more targeted, specialized, refined, digitalized, and intelligentized. Key influencing factors are selected as reference data for analysis in various evaluation methods, as shown in Fig. 7.

Each evaluation method has certain limitations, as shown in Table 1. In this paper, the following conclusions are drawn based on the analysis:

1. In terms of application, the qualitative and quantitative evaluation methods are suitable for the evaluation of ports' comprehensive competitiveness, and the deductive and retrospective evaluation methods are suitable for dominance evaluation of a certain aspect or factor of ports.
2. In terms of evaluation objects, with the development of information technology, the port dominance evaluation methods have changed from comprehensive analysis to evaluating a key influencing factor. This implies that researchers focus on pertinence, refinement and locality of the evaluation objects and the evaluation objects are targeted, refined, and localized.
3. In terms of the evaluation method, it changes from the evaluation of ports' comprehensive competitiveness using key influencing factors microscopically to the detection of weaknesses and risk points of key influencing factors based on the macroscopic manifestation of ports' comprehensive competitiveness.
4. In terms of reference elements, with the development of advanced intelligent technologies such as big data and artificial intelligence, evaluation methods are more and more dependent on data and intelligent algorithms, which require higher and higher computing power of computers.

4. Problems and challenges

Ports, as a factor driving the regional economic growth, influence the region's overall competitiveness and further affect the national competitiveness and international status. This article summarizes the methods used to evaluate port dominance that influences the port competitiveness. The port dominance is a complex comprehensive evaluation result, which is directly affected by various factors and evaluation methods. Through the analysis of existing evaluation methods, some problems are found as follows:

1. The evaluation of port dominance involves multiple factors and uncertain conditions. Most of the evaluation methods aim to evaluate port dominance with key influencing factors, which will lead to single-faceted results.
2. With the development of the port and surrounding environment, port dominance will change accordingly. Therefore, dominance will vary with the change of dynamic influencing factors, which should be taken into account in the evaluation method of port dominance.
3. Due to the complexity of ports, it is difficult to use a specific method and technology for reasonable evaluation. To provide port dominance and similarity analysis for shipping networks, knowledge

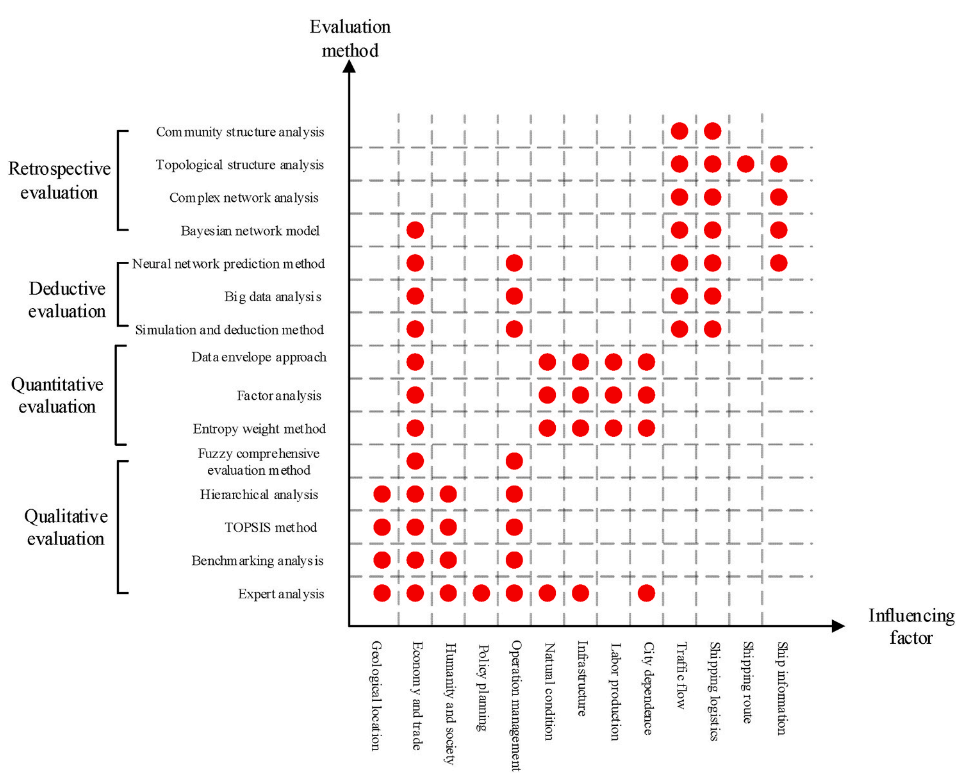

Fig. 7. Various evaluation methods.

Table 1
Comparison of evaluation methods.

Method	Arithmetic	Application condition	Advantage	Disadvantage
Qualitative evaluation methods	Expert analysis	Evaluation of the comprehensive competitiveness	The method is simple and easy to use	Subject to personal subjective factors
	Benchmarking analysis	The benchmark ports are compared and analyzed	The method is simple and easy to use	Limited application
	TOPSIS method	The relative proximity degree is used to rank the port dominance	The method is simple and easy to use	Limited application
	Hierarchical analysis	Rank the port dominance based on different evaluation dimensions and evaluation index systems	The method is simple and easy to use	Limited application
	Fuzzy comprehensive evaluation method	A fuzzy judgment matrix is built to make a comprehensive evaluation of uncertain factors in evaluating port dominance using the fuzzy mathematical method	Conducive to the analysis of uncertain factors	Modeling analysis and complex algorithms are required
Quantitative evaluation methods	Entropy weight method	The entropy weight is used as the weight value to objectively evaluate many factors affecting port dominance	Objective analysis	Heavy weight and hard quantification
	Factor analysis	The factors greatly affecting port dominance are fused into one factor, and only several factors are used to analyze and evaluate port dominance.	Objective analysis	Key factors are not easily quantified
	Data envelope approach	By using the optimal weight to calculate the efficiency value of other influencing factors, the most critical factors affecting the port comprehensive efficiency can be found	Easy to find the most critical factors	Too much calculation is required
Evaluation methods	Simulation and deduction method	The simulation and deduction method is to find the significant nonlinear characteristics through the statistics of ship traffic flow data, and to deduce the future development trend through the modeling of historical data	Quantified factor and strong data pertinence	Highly difficult, large computation and complex algorithms are required
	Big data analysis and prediction method	Through the accumulation of historical data, with the help of big data analysis, the internal relations and laws between the data are mined	Quantified factor, accurate forecast, strong data pertinence	Highly difficult, too much calculation, complex algorithms, and large amount of data are required
	Neural network reasoning method	Port future development reasoning and forecasting applications	Quantified factor, accurate prediction, relatively new technology	Too much calculation and complex algorithms are required
Retrospective evaluation methods	Bayesian network model	The degree of shipping patency between ports and the degree of trade intimacy between ports can be accurately calculated. The high-risk factors that affect port operation efficiency and shipping patency rate are analyzed	Quantified factor, accurate calculation	Too much calculation and complex algorithms are required
	Complex network analysis	The role of ports in the region is demonstrated from a different perspective	Macroscopic evaluation, and accurate prediction	Too much calculation and complex algorithms are required.
	Topological structure analysis	Ports are evaluated from the perspective of shipping network analysis connectivity and functional robustness	Macroscopic evaluation and accurate prediction of key ports	Too much calculation and complex algorithms are required.
	Community structure analysis	Analyze the different positions of ports or groups of ports in global networks and community networks. The advantages of ports are complementary, which can be used as a method to evaluate the superiority of ports in the region.	Macroscopic evaluation and accurate prediction of key ports	Too much calculation and complex algorithms are required.

mapping, for example, can be used to superpose heterogeneous information such as port attributes, arrival records in ports, and government background. Therefore, how to effectively take into account different methods, including knowledge mapping, is a direction of academic development of shipping networks, so as to provide a more accurate basis for port evaluation.

The analysis shows that the port involves many influencing factors with subtle correlation, which restrict and influence each other. In addition, the port is known as a dynamic development system. Thus, the port's continuous changes, and other factors such as peripheral environment, social and technological progress, and interest of shipping companies, such as berths, channels, even hinterland, and road infrastructure cause the port's advantage to change continuously so that the advantage is difficult to be characterized. Therefore, a combination of various evaluation methods is needed to evaluate port dominance. Note that the influencing factors related to the port's dominance can efficiently present the port's real competitiveness.

5. Proposals and prospects

The competition among ports has intensified with the diversified development of the world. Also, factors that affect port dominance involve multiple fields and disciplines. These influencing factors are changing with the development of modern information technology, the innovation of business models adopted by ports, the development trends of global trade, and the transformation and upgradation of national economic strategies. Hence, the difficulty of port competitiveness evaluation has increased. Based on the review of relevant literature and understanding of this field, the research direction of port dominance evaluation method is proposed as follows:

1. The comprehensive evaluation of port competitiveness could be carried out by integrating various networks. In the study of shipping networks, a port is a node of a complex network. Meanwhile, the port integrates a variety of complex networks, such as the business network, communication network, monitoring network, and management network. Also, a variety of external complex networks affect the port, including highway networks, railway networks, aviation networks, power networks, communication networks, Internet, biological networks, and social networks, which directly or indirectly affect ports' comprehensive competitiveness. Note that the research on ports' competitiveness from the perspective of the integration of various networks inside and outside the ports is limited.
2. Port intelligent management and unmanned operation have become a trend for improving the efficiency. For example, Yang Shan Port has become the first intelligent unmanned operation port in China. Big data and artificial intelligence technology have become an important means to improve port operation efficiency. In the future, with the development and use of 5G, smart cars, smart ships, and

other advanced technologies, the competitiveness of ports will be greatly improved, and the degree of intelligence and unmanned operation will be an important factor to measure port dominance. Therefore, the research on intelligent standards in port dominance evaluation will be further developed.

3. Multi-agent comprehensive evaluation of port dominance has gradually developed. The port undertakes functions such as sea-land transport, internal and external trade, vehicle and ship exchange, involving local governments, customs, coast guards, navy, health departments, maritime traffic, maritime affairs, aquaculture practitioners, freight forwarders, shipping agencies, shipping companies, and many other behavioral agents. The comprehensive performance of these agents reflects the competitiveness of ports and will become an important factor affecting the evaluation of port dominance. However, there are few studies on multi-agent comprehensive evaluation of port dominance.
4. Ecological environmental protection should be taken as an important index in the evaluation of port dominance. The increasing frequency of shipping trade has caused environmental pollution such as carbon emissions, low-sulfur oil, ballast water, and oil leakage. These pollution have caused damage to the ecological environment of the sea and land areas surrounding the port. Some major pollution incidents directly affect the normal operation of the port. Note that promoting a comprehensive and green transformation of the economy and society with a high level of environmental protection has become an important goal of China's economic development at this stage. However, the research on the ecological environment as one of the indices of port dominance is in its initial stage.

Author contributions

Conceptualization, P.W. and Q.H.; methodology, F.W.; software, Q.M.; formal analysis, Y.X.; investigation, Y.X.; resources, Q.M.; writing—original draft preparation, P.W.; writing—review and editing, F.W.; visualization, Y.X.; supervision, F.W.; project administration, Q.H.; funding acquisition, Q.H. All authors have read and agreed to the published version of the manuscript.

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Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests: Peng Wang reports was provided by Institute of Computing Technology Chinese Academy of Sciences. Peng Wang reports a relationship with Institute of Computing Technology Chinese Academy of Sciences that includes: employment.

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